L'HÔPITAL'S RULE

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Name: Solutions

1. Compute

$$\lim_{x\to 1} \frac{x^3 - 2x^2 + 1}{x^3 - 1}$$

in two ways: with and without using L'Hôpital's Rule.

$$\lim_{x\to 1} \frac{x^3-2x^2+1}{x^3-1} \overset{L'H}{=} \lim_{x\to 1} \frac{3x^2-4x}{3x^2} = \frac{3-4}{3} = \boxed{\frac{-1}{3}}$$

Without L'Hôpital: factor an $x-1$ out of each term

$$x^3-1 = (x-1)(x^2+x+1)$$

$$\begin{array}{r|l} & x^2 - x - 1 \\ \hline x-1 & x^3 - 2x^2 + 1 \\ & -x^3 + x^2 \\ \hline & -x^2 + 1 \\ & +x^2 - x \\ \hline & -x + 1 \\ & +x - 1 \\ \hline & 0 \end{array} \Rightarrow \lim_{x\to 1} \frac{x^3-2x^2+1}{x^3-1} = \lim_{x\to 1} \frac{(x-1)(x^2-x-1)}{(x-1)(x^2+x+1)}$$

$$= \lim_{x\to 1} \frac{x^2-x-1}{x^2+x+1}$$

$$= \frac{1-1-1}{1+1+1}$$

$$= \boxed{\frac{-1}{3}}$$

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Evaluate the following limits.

2. $\lim_{x\to\pi} \frac{\sin(3x)}{x-\pi}$

$\overset{L'H}{=} \lim_{x\to\pi} \frac{3\cos(3x)}{1}$

$= \boxed{3}$

3. $\lim_{t\to 0} \frac{e^{2t}-1}{e^t}$

$= \frac{e^0-1}{e^0}$

$= \frac{1-1}{1}$

$= \boxed{0}$

4. $\lim_{\theta\to 0} \frac{\arctan(\theta)}{2\theta}$

$\overset{L'H}{=} \lim_{t\to 0} \frac{\frac{1}{1+\theta^2}}{2}$

$= \frac{1}{2}\left(\frac{1}{1+0}\right)$

$= \boxed{\frac{1}{2}}$

5. $\lim_{x\to\infty} \frac{e^{-x}}{1+\ln(x)}$

$= \lim_{x\to\infty} \frac{1}{e^x(1+\ln(x))}$

$= \boxed{0}$

because

$e^x(1+\ln(x)) \to \infty$

as $x \to \infty$.

6. $\lim_{x\to\infty} \frac{(\ln(x))^2}{x}$

$\overset{L'H}{=} \lim_{x\to\infty} \frac{2\ln(x)(\frac{1}{x})}{1}$

$= \lim_{x\to\infty} \frac{2\ln(x)}{x}$

$\overset{L'H}{=} \lim_{x\to\infty} \frac{2(\frac{1}{x})}{1}$

$= \lim_{x\to\infty} \frac{2}{x}$

$= \boxed{0}$

7. $\lim_{u\to\infty} \frac{\sqrt{u^2+1}}{u}$

$= \lim_{u\to\infty} \frac{\sqrt{u^2}\sqrt{1+1/u^2}}{u}$ $(0<u)$

$= \lim_{u\to\infty} \sqrt{1+1/u^2}$

$= \sqrt{\lim_{u\to\infty} 1+1/u^2}$

$= \sqrt{1+0}$

$= \boxed{1}$

8. $\lim_{x\to\infty} (x - \ln(x))$

$$= \lim_{x\to\infty} \ln(e^x) - \ln(x)$$

$$= \lim_{x\to\infty} \ln\left(\frac{e^x}{x}\right)$$

$$= \boxed{\infty}$$

because

$$\lim_{x\to\infty} \frac{e^x}{x} \overset{L'H}{=} \lim_{x\to\infty} \frac{e^x}{1} = \infty$$

and $\ln(x) \to \infty$ as $x \to \infty$.

9. $\lim_{x\to 1} \left(\frac{x}{x-1} - \frac{1}{\ln(x)}\right)$

$$= \lim_{x\to 1} \frac{x\ln(x) - x + 1}{(x-1)\ln(x)}$$

$$\overset{L'H}{=} \lim_{x\to 1} \frac{\ln(x) + 1 - 1}{\ln(x) + (x-1)/x}$$

$$= \lim_{x\to 1} \frac{\ln(x)}{\ln(x) + 1 - 1/x}$$

$$\overset{L'H}{=} \lim_{x\to 1} \frac{\frac{1}{x}}{\frac{1}{x} + \frac{1}{x^2}}$$

$$= \frac{1}{1+1}$$

$$= \frac{1}{2}$$

10. $\lim_{x\to 1^+} \left[\ln(x^7 - 1) - \ln(x^5 - 1)\right]$

$$= \lim_{x\to 1^+} \ln\left(\frac{x^7 - 1}{x^5 - 1}\right)$$

$$= \ln\left(\lim_{x\to 1^+} \frac{x^7 - 1}{x^5 - 1}\right)$$

$$\overset{L'H}{=} \ln\left(\lim_{x\to 1^+} \frac{7x^6}{5x^6}\right)$$

$$= \boxed{\ln(7/5)}$$

11. $\lim_{x\to 0^+} x^{\sqrt{x}}$

$$\ln\left(\lim_{x\to 0^+} x^{\sqrt{x}}\right)$$

$$= \lim_{x\to 0^+} \ln\left(x^{\sqrt{x}}\right)$$

$$= \lim_{x\to 0^+} \sqrt{x}\ln(x)$$

$$= \lim_{x\to 0^+} \frac{\ln(x)}{x^{-1/2}}$$

$$\overset{L'H}{=} \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{2}x^{-3/2}}$$

$$= \lim_{x\to 0^+} \frac{-2x^{3/2}}{x}$$

$$= \lim_{x\to 0^+} -2x^{1/2}$$

$$= 0$$

$$\Rightarrow \lim_{x\to 0^+} x^{\sqrt{x}} = e^0 = \boxed{1}$$

12. $\lim_{x\to 0} (1 - 2x)^{1/x}$

$$\lim_{x\to 0} \frac{1}{x}\ln(1-2x)$$

$$\overset{L'H}{=} \lim_{x\to 0} \frac{\frac{-2}{1-2x}}{1}$$

$$= \frac{-2}{1-0}$$

$$= -2$$

$$\Rightarrow \lim_{x\to 0} (1-2x)^{1/x} = \boxed{e^{-2}}$$

13. $\lim_{x\to\infty} x^{1/x}$

$$\lim_{x\to\infty} \frac{\ln(x)}{x} \overset{L'H}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{1}$$

$$= 0$$

$$\Rightarrow \lim_{x\to\infty} x^{1/x} = e^0 = \boxed{1}$$